

Dual band. Dual-band wireless-N gigabit routers such as Linksys by Cisco WRT610N, Netgear's WNR3500 and TRENDnet's 300Mbps Dual-band Wireless-N Gigabit Router enable the user to take advantage of 802.11n grade wireless connections operating in the 2.4GHz wireless band and/or the wider 5GHz band. The 5GHz band is less crowded than the 2.4GHz band that is used by other devices like DECT phones or microwaves, and therefore better suited for carrying latency sensitive data like video and audio. A dual-band wireless router is like having two N band routers in one device. "The 5GHz band is ideal for streaming media and provides more than seven times the number of non-overlapping wireless channels, allowing for clearer transmissions. The 2.4GHz spectrum allows older network devices like print servers, cameras or older notebooks and PCs to access the network. As opposed to other solutions, the WRT610N is designed to allow consumers to achieve full Wireless-N network speeds on both bands at the same time," explains Balasubramanian.

Push 'n' connect. Some of the newer wireless-N routers such as Netgear's WNR3500 RangeMax Wireless-N Gigabit Router provide a feature called 'push 'n' connect'. This feature basically allows you to wirelessly connect to supported devices easily without having to remember the security password and enter it every time. 'Push 'n' connect' is based on the Wi-Fi Protected Setup (WPS) standard. It enables you to configure the router's network name (SSID) and security settings and then easily connect wireless clients securely and easily to the router. WPS automatically configures the network name (SSID) and wireless security settings for the router (if the router is in its default state) and broadcasts these settings to the wireless client. So, you do not have to remember cumbersome security settings and details every time you want to connect a wireless client.

The future of wireless routers is all about integration—solutions that are highly secure, flexible and built to be compatible with future technologies

Powerline AV adapter. A wireless networking related innovation that won an innovation award at this year's Consumer Electronics Show (CES) is Netgear's Powerline AV Adapter with an integrated four-port fast ethernet switch. The adapter turns any electrical power outlet into an Ethernet network connection that can handle up to 200 Mbps speed. According to them, this 'no new wires' technology will allow users to carry high-speed data network traffic on existing electrical wiring. The adapter is designed for use with high-performance applications like high-definition video streaming, networked attached storage and console gaming.

M-Drive technology. M-Drive is a significant innovation from Cisco, which is known for having come out with the first enterprise-class unified wireless network platform using the full potential of 802.11n. M-Drive technology is a new system-wide set of features that further enhances 802.11n performance by improving wireless coverage and capacity, optimising device connections and simplifying wireless management. One particular feature of M-Drive technology, ClientLink (yet to be launched) also helps extend the life of existing 802.11a and g devices. The technology uses beam forming to improve the throughput of existing 802.11a and g devices, increase overall wireless channel capacity and reduce wireless coverage holes for legacy devices.

Xtreme N. D-Link's award-winning Xtreme N product family has several interesting products that tap the potential of 802.11n technology, including the GamerLounge and the

MediaLounge. Their new product, the DIR 685 all-in-one router won an innovation award at CES 2009. The all-in-one router is a wireless-N router that doubles up as a network attached storage (NAS) device. It also has a 8.1cm (3.2-inch) LCD monitor for displaying photos, desktop applications and network performance diagnostics. Truly an all-in-one device!

The future is all about integration

The future of wireless routers is all about integration—solutions that are highly secure, flexible and built to be compatible with future technologies. In a way, such integrated network router solutions are already here, and the trend is likely to strengthen in the coming years.

We will see the router transform into an intelligent device that performs multiple functions including prioritisation of network applications—a feature that is already here, thanks to 802.11n. With high speed, reliable transmission, routers will easily bring the office right to the worker's living room—telecommuting will be much easier than it is now. Integrated services will enable technical staff to monitor the health and security of devices from a central location. Intelligent routers and other network devices will make it possible for the network itself to take up the responsibility for its security at various levels, rather than relying on a centralised software-based security solution. This will help protect companies from the influx of viruses, malicious code and other infections that end users' laptops might unknowingly acquire.

In short, a network will no longer be a mere collection of devices but a well-rehearsed ballet performed by intelligent devices. No more disruptions, no more fears. We are at the brink of an age of fast, secure, integrated wireless networks. ●

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